UNIT – III: Electromagnetic Theory in Quantum Context

Full step-by-step derivations • Full vector calculus proofs • TE/TM full derivations • Cavity quantization via Hamiltonian • 25 solved university exam problems • Classical vs Quantum comparison

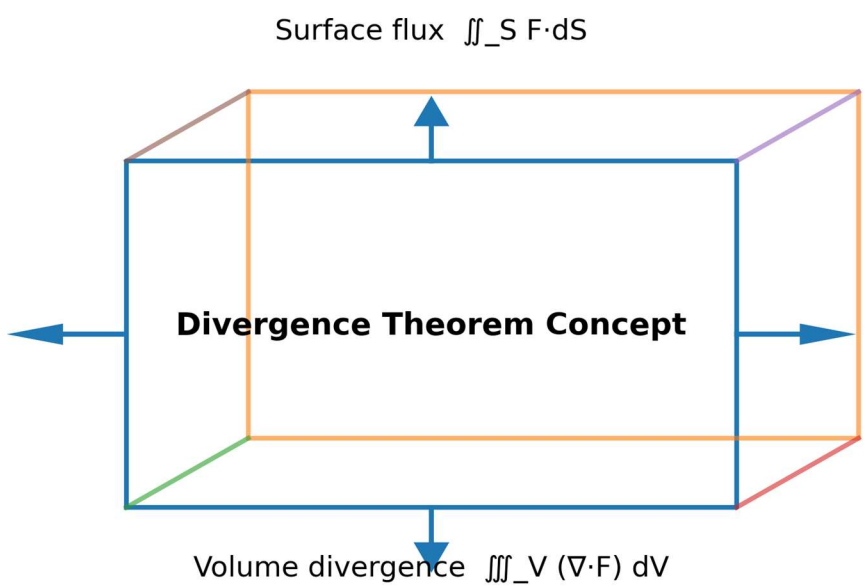
This is a textbook-style chapter for ECE/Quantum Technologies courses.

1. Vector Calculus Toolkit for Electromagnetics (Full Proof-Style Notes)

1.1 Divergence Theorem (Gauss Theorem) — Proof Outline

Statement: $\iiint_V (\nabla\cdot F)\, dV = \iint_S F\cdot dS$, where $S = \partial V$.

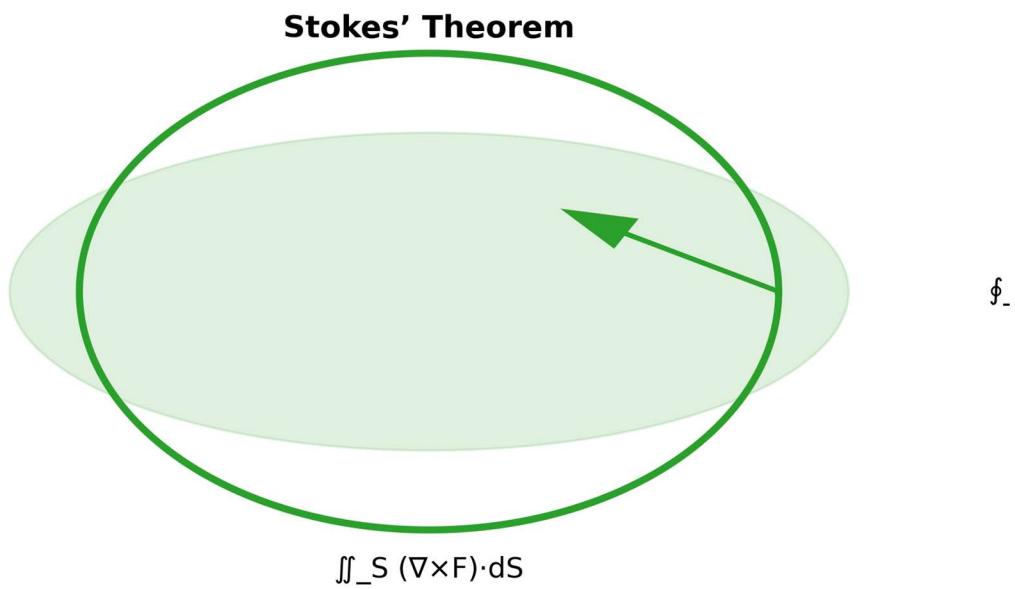
Proof (small rectangular box): For a box $\Delta x\Delta y\Delta z$, flux through opposite x-faces is $[F_x(x+\Delta x/2)-F_x(x-\Delta x/2)]\Delta y\Delta z \approx (\partial F_x/\partial x)\Delta x\Delta y\Delta z$. Similarly y and z contributions give net outward flux $\approx (\nabla\cdot F)\Delta V$. Summing over all boxes cancels internal faces, leaving only boundary flux; limit gives the theorem.



1.2 Stokes' Theorem — Proof Outline

Statement: $\iint_S (\nabla\times F)\cdot dS = \oint_C F\cdot dl$, where $C=\partial S$.

Proof (tiling): Divide S into small patches. Circulation around each patch boundary $\approx (\nabla\times F)\cdot n\, \Delta S$. Interior edges cancel in the sum because adjacent loops traverse them in opposite directions; remaining boundary is C.



1.3 Identity: curl(curl(A))

Identity: $\nabla\times(\nabla\times A) = \nabla(\nabla\cdot A) - \nabla^2 A$.

Component sketch using Levi-Civita symbols yields $(\nabla\times(\nabla\times A))_i = \partial/\partial x_i(\nabla\cdot A) - \nabla^2 A_i$.

2. Maxwell's Equations → Wave Equation → Helmholtz Equation (No Skipped Steps)

2.1 Maxwell's Equations (Differential Form)

$\nabla\cdot D = \rho_v$, $\nabla\cdot B = 0$, $\nabla\times E = -\partial B/\partial t$, $\nabla\times H = J + \partial D/\partial t$.

Constitutive: $D=\varepsilon E$, $B=\mu H$, $J=\sigma E$ (ohmic) in linear isotropic media.

2.2 Derivation of Vector Wave Equation for E (source-free homogeneous medium)

Assume $\rho_v=0$ and $J=0$ inside the medium; ε and μ are constants.

Step 1: Faraday: $\nabla\times E = -\partial B/\partial t = -\mu\, \partial H/\partial t$.

Step 2: Take curl: $\nabla\times(\nabla\times E) = -\mu\, \partial/\partial t(\nabla\times H)$.

Step 3: Ampere–Maxwell ($J=0$): $\nabla\times H = \partial D/\partial t = \varepsilon\, \partial E/\partial t$.

Step 4: Substitute: $\nabla\times(\nabla\times E) = -\mu\, \partial/\partial t(\varepsilon\, \partial E/\partial t)=-\mu\varepsilon\, \partial^2 E/\partial t^2$.

Step 5: Use identity: $\nabla\times(\nabla\times E)=\nabla(\nabla\cdot E)-\nabla^2 E$.

Step 6: From $\nabla\cdot D=0 \Rightarrow \nabla\cdot(\varepsilon E)=0 \Rightarrow \nabla\cdot E=0$ (ε constant).

Step 7: Hence $\nabla(\nabla\cdot E)=0$ and we get $\nabla^2 E = \mu\varepsilon\, \partial^2 E/\partial t^2$.

2.3 Helmholtz equation

Let $E(r,t)=\mathrm{Re}\{E(r)\, e^{j\omega t}\}$. Then $\partial^2/\partial t^2 \to -\omega^2$.

So $\nabla^2 E + k^2 E = 0$, where $k = \omega\sqrt{(\mu\varepsilon)}$.

3. Lossy Media: Complex Permittivity and Propagation Constant

In a conductor: $J=\sigma E$, so $\nabla\times H = \sigma E + \varepsilon\partial E/\partial t$.

Phasor: $\nabla\times H=(\sigma+j\omega\varepsilon)E$ and $\nabla\times E=-j\omega\mu H$.

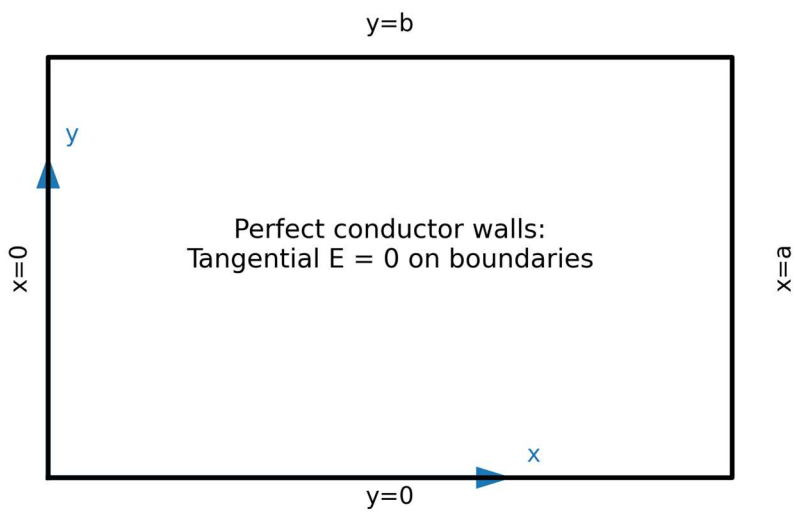
Curl Faraday: $\nabla\times(\nabla\times E)=-j\omega\mu(\nabla\times H)=-j\omega\mu(\sigma+j\omega\varepsilon)E$.

Assuming $\nabla\cdot E\approx 0$: $-\nabla^2 E = -j\omega\mu(\sigma+j\omega\varepsilon)E \Rightarrow \nabla^2 E = j\omega\mu(\sigma+j\omega\varepsilon)E$.

For plane wave: $d^2E/dz^2 = \gamma^2 E$, $\gamma=\sqrt{(j\omega\mu(\sigma+j\omega\varepsilon))}=\alpha+j\beta$.

Good conductor: $\gamma\approx(1+j)\sqrt{(\omega\mu\sigma/2)} \Rightarrow \alpha\approx\beta\approx\sqrt{(\pi f \mu \sigma)}$, skin depth $\delta=1/\alpha=\sqrt{(2/(\omega\mu\sigma))}$.

4. Rectangular Waveguide: Full TE and TM Mode Derivations



TE modes

$E_z = 0, \ H_z \neq 0$

TM modes

$H_z = 0, \ E_z \neq 0$

TEM mode

$E_z = H_z = 0$ (needs two conductors)

Mode classification by longitudinal components

4.1 Separation of fields into transverse/longitudinal

Assume fields vary as $e^{j\omega t}e^{-j\beta z}$. Write $E=E_t+\hat{z}E_z$ and $H=H_t+\hat{z}H_z$.

Define transverse Laplacian: $\nabla_t^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$.

4.2 Transverse Helmholtz equations

Longitudinal components satisfy: $(\nabla_t^2 + k_c^2)E_z = 0$ and $(\nabla_t^2 + k_c^2)H_z = 0$.

With $k_c^2 = k^2 - \beta^2$ and $k=\omega\sqrt{(\mu\varepsilon)}$. Hence $\beta = \sqrt{(k^2 - k_c^2)}$.

4.3 TE modes (E_z=0): determine H_z with Neumann boundary conditions

TE: $E_z=0$, $H_z\neq0$. Perfect conductor implies tangential $E=0$ on walls, giving $\partial H_z/\partial n=0$.

Solve $(\nabla_t^2 + k_c^2)H_z=0$ with $\partial H_z/\partial x=0$ at $x=0,a$ and $\partial H_z/\partial y=0$ at $y=0,b$.

Separation: $H_z=X(x)Y(y)$. Choose $X=\cos(m\pi x/a)$, $Y=\cos(n\pi y/b)$.

Then $k_c^2=(m\pi/a)^2+(n\pi/b)^2$ with $m,n=0,1,2,...$ (not both zero).

4.4 TM modes (H_z=0): determine E_z with Dirichlet boundary conditions (FULL)

TM: $H_z=0$, $E_z\neq0$. For perfect conductor walls, E_z must vanish on boundaries: $E_z=0$ at $x=0,a$ and $y=0,b$.

Solve $(\nabla_t^2 + k_c^2)E_z=0$ with Dirichlet BCs.

Separation: $E_z=X(x)Y(y)$ with $X=\sin(m\pi x/a)$, $Y=\sin(n\pi y/b)$, $m,n=1,2,...$

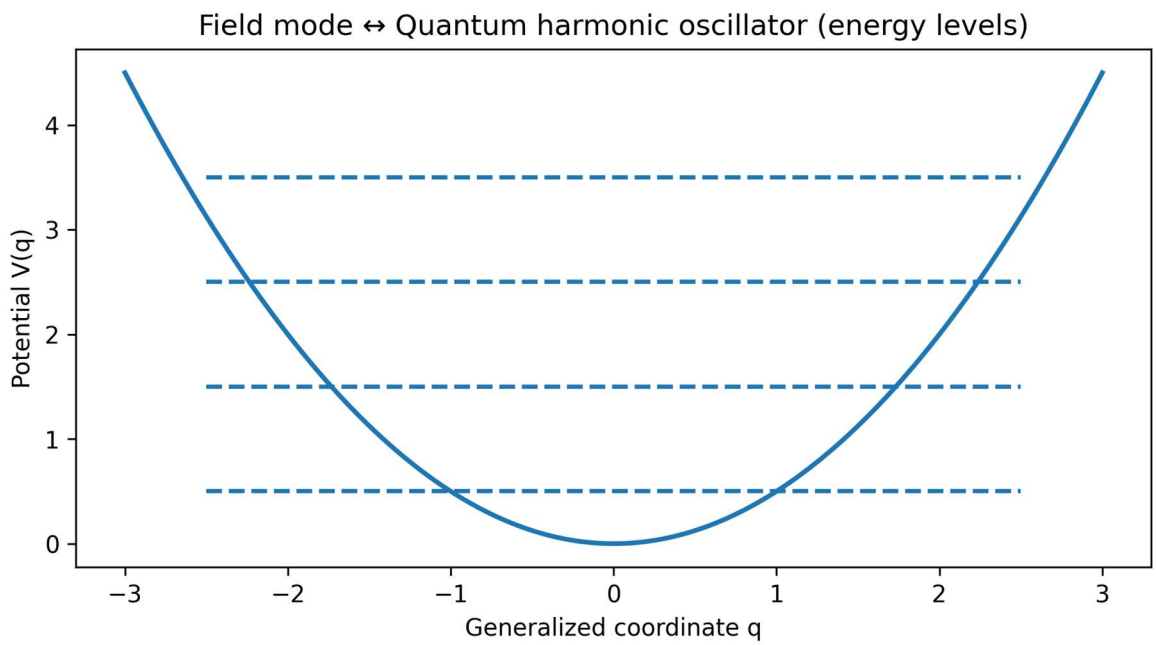
Then $k_c^2=(m\pi/a)^2+(n\pi/b)^2$ and $\beta=\sqrt{(k^2-k_c^2)}$.

4.5 Cutoff frequency and dispersion

Cutoff: $\beta=0 \Rightarrow k=k_c$. For air-filled: $f_c=(c/2)\sqrt{((m/a)^2+(n/b)^2)}$. Dominant mode: TE10 with $f_c=c/(2a)$.

Phase velocity $v_p=\omega/\beta$ (>c); group velocity $v_g=d\omega/d\beta=c^2/v_p$ (<c) for vacuum guide.

5. Cavity Quantization using Hamiltonian Formalism



5.1 Classical field energy

$U = \iiint_V [(1/2)\varepsilon|E|^2 + (1/2)\mu|H|^2] dV$.

5.2 Mode expansion and reduction to harmonic oscillators

In a cavity, boundary conditions yield discrete modes with frequencies ω_λ.

Expand fields in modes so total energy becomes: $H = \Sigma_\lambda [(1/2)p_\lambda^2 + (1/2)\omega_\lambda^2 q_\lambda^2]$.

5.3 Canonical quantization and ladder operators (step-by-step)

Promote q_λ, p_λ to operators with $[\hat{q}_\lambda, \hat{p}_{\lambda'}]=i\hbar\, \delta_{\lambda\lambda'}$.

Define $\hat{a}_\lambda = \sqrt{(\omega_\lambda/(2\hbar))}\, \hat{q}_\lambda + i\sqrt{(1/(2\hbar\omega_\lambda))}\, \hat{p}_\lambda$ and $\hat{a}_\lambda^\dagger$ accordingly.

Then $H = \Sigma_\lambda \hbar\omega_\lambda (\hat{a}_\lambda^\dagger \hat{a}_\lambda + 1/2)$. Photon number $\hat{n}=\hat{a}^\dagger\hat{a}$.

Energy eigenvalues: $E_n = \hbar\omega(n+1/2)$ and spacing $\Delta E=\hbar\omega=hf$.

6. Classical EM vs Quantum Electromagnetic Field (Comparison)

Classical EM: deterministic continuous fields E,H. Quantum field: operator-valued fields with quantized excitations (photons).

Comparison points:
• Variables: $E(r,t), H(r,t)$ vs $\hat{E}(r), \hat{H}(r)$
• Energy: U continuous vs $U=\Sigma \hbar\omega(n+1/2)$
• Measurement: amplitude/phase vs photon counts/correlations ($g^{(2)}$, antibunching)
• Noise: thermal/electronic vs vacuum fluctuations (zero-point)
• States: coherent waves vs quantum states (Fock, coherent, squeezed, entangled)
• Boundary conditions/modes: Maxwell still determines mode shapes in both pictures.

7. Solved University Exam Problems (25) — Worked

Problem 1
Task: Derive wave equation from Maxwell's equations

Solution: Using $\nabla\times E=-\partial B/\partial t$ and $\nabla\times H=\partial D/\partial t$, take curl of Faraday, substitute Ampere–Maxwell, use $\nabla\times(\nabla\times E)=\nabla(\nabla\cdot E)-\nabla^2 E$ and $\nabla\cdot E=0$ to get $\nabla^2 E=\mu\varepsilon\partial^2 E/\partial t^2$.

Problem 2
Task: Wavenumber in dielectric

Solution: $f=3$ GHz, $\varepsilon_r=2.25$, $\mu_r=1$: $k=(2\pi f/c)\sqrt{(\varepsilon_r\mu_r)}= (2\pi\cdot 3e9/3e8)\cdot 1.5 = 94.25$ rad/m.

Problem 3
Task: Intrinsic impedance

Solution: $\varepsilon_r=4$: $\eta=\eta 0/\sqrt{\varepsilon_r}=377/2=188.5\ \Omega$.

Problem 4
Task: Wavelength and β

Solution: $\varepsilon_r=9$, $f=2$ GHz: $v=c/3=1e8$; $\lambda=v/f=0.05$ m; $\beta=2\pi/\lambda=125.66$ rad/m.

Problem 5
Task: TE10 cutoff

Solution: $a=2$ cm: $f_c=c/(2a)=3e8/0.04=7.5$ GHz.

Problem 6
Task: TM11 cutoff

Solution: $a=3$ cm, $b=2$ cm: $f_c=(c/2)\sqrt{((1/a)^2+(1/b)^2)}=9.01$ GHz.

Problem 7
Task: β above cutoff

Solution: TE10, $a=2.5$ cm, $f=10$ GHz: $k=209.44$, $k_c=\pi/a=125.66$, $\beta=167.5$ rad/m.

Problem 8
Task: Guide wavelength

Solution: $\lambda_g=2\pi/\beta=0.0375$ m (from $\beta=167.5$).

Problem 9

Task: Group velocity

Solution: At $f=2f_c$: $v_g=c\sqrt{(1-(f_c/f)^2)}=0.866c$.

Problem 10

Task: Phase velocity

Solution: At $f=2f_c$: $v_p=c/\sqrt{(1-(f_c/f)^2)}=1.155c$.

Problem 11

Task: Skin depth copper 1 MHz

Solution: $\delta=\sqrt{(2/(\omega\mu\sigma))}$ with $\omega=2\pi\cdot 1e6$, $\mu=\mu_0$, $\sigma=5.8e7 \Rightarrow \delta\approx 66$ μm.

Problem 12

Task: Skin depth copper 10 MHz

Solution: δ scales as $1/\sqrt{f} \Rightarrow 66/\sqrt{10}\approx 20.9$ μm.

Problem 13

Task: Loss tangent

Solution: $\varepsilon_r=4$, $\sigma=0.01$, $f=1$ GHz: $\tan\delta=\sigma/(\omega\varepsilon)=0.0449$.

Problem 14

Task: Photon energy at 1550 nm

Solution: $E=hc/\lambda=1.28\times 10^{-19}$ J ≈ 0.80 eV.

Problem 15

Task: Photon energy at 5×10^{14} Hz

Solution: $E=hf=3.313\times 10^{-19}$ J ≈ 2.07 eV.

Problem 16

Task: Average power density

Solution: Free space, $E_0=200$ V/m: $\langle S\rangle=E_0^2/(2\eta_0)=53.05$ W/m^2.

Problem 17

Task: Energy density

Solution: Free space, $E_{rms}=100$ V/m: $\langle u\rangle=\varepsilon_0 E_{rms}^2=8.854\times 10^{-8}$ J/m^3.

Problem 18

Task: Cavity fundamental frequency

Solution: $L=5$ cm: $f_1=c/(2L)=3$ GHz.

Problem 19

Task: Cavity energy spacing

Solution: $\Delta E=hf=(6.626e-34)(3e9)=1.99e-24$ J.

Problem 20

Task: Fiber NA

Solution: $n_1=1.48$, $n_2=1.46$: NA=0.242; $\theta_{max}\approx 14°$.

Problem 21

Task: Cutoff wavenumber

Solution: $k_c^2=(m\pi/a)^2+(n\pi/b)^2$ from separation of variables and BCs.

Problem 22

Task: TE boundary condition

Solution: For TE modes in rectangular guide: $\partial H_z/\partial n=0$ at conducting walls; leads to cosine solutions.

Problem 23

Task: TM boundary condition

Solution: For TM modes: $E_z=0$ at conducting walls; leads to sine solutions.

Problem 24

Task: Compute f_c for TE20

Solution: TE20: $m=2, n=0 \Rightarrow f_c=c/a$. Example $a=2.5$ cm $\Rightarrow f_c=12$ GHz.

Problem 25

Task: Compute β for TM11 example

Solution: Given $a=3$ cm, $b=2$ cm, $f=12$ GHz: $k=2\pi f/c=251.33$;
$k_c=\pi\sqrt{(1/a)^2+(1/b)^2}=188.73$; $\beta=\sqrt{k^2-k_c^2}=165.3$ rad/m.

8. Quick Revision Sheet

Wave: $\nabla^2 E=\mu\varepsilon\partial^2 E/\partial t^2$; Helmholtz: $\nabla^2 E+k^2 E=0$, $k=\omega\sqrt{\mu\varepsilon}$.

Lossy: $\gamma=\sqrt{j\omega\mu(\sigma+j\omega\varepsilon)}=\alpha+j\beta$; $\delta=\sqrt{2/(\omega\mu\sigma)}$.

Waveguide: $k_c^2=(m\pi/a)^2+(n\pi/b)^2$; $\beta=\sqrt{k^2-k_c^2}$; $f_c=(c/2)\sqrt{(m/a)^2+(n/b)^2}$.

Quantization: $H=\Sigma\,\hbar\omega(a^\dagger a+1/2)$, $\Delta E=\hbar\omega=hf$.